Computable Countermarkets and the Limits of Universal Trading

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We explain why no trading algorithm can guarantee profit in every market. For each deterministic program that always returns an exact rational position, we construct a fixed, algorithmically generated price path on which every active position loses and inactivity earns nothing. This holds with positive, continually changing prices, costless trading, and unlimited computation time. Conversely, a supplied total computable price program with nonzero moves admits a computable trader that profits at every step. Mutual countering depends on information, timing, and termination. Approximation-based real outputs preserve the cumulative obstruction but not the loss on every active position. Separate arguments limit learning market rules, certifying future events, and establishing randomness from finite data. Betting-martingale characterizations connect trading with computable and Martin-Löf randomness. Useful strategies may exploit market structure, information, or compensation for risk, while benchmark performance need not imply profit. Reversing and rearranging price histories within the assumed market class provide practical stress tests, distinguishing conditional success from universal guarantees.

I. INTRODUCTION

Throughout this paper the elementary configuration has two sides: a *trader* and a *market*. At each date the trader converts observed history into a position, meaning its chosen asset holdings; the market supplies the next price. Depending on the question, the market side is a realized path (a sequence of prices), a path-generating program, a probability law assigning probabilities to possible outcomes, or a price process constrained by no-arbitrage. A price process models prices as random variables over time; no-arbitrage excludes a risk-free opportunity to gain from zero initial capital [1, 2]. Holding the trader fixed while changing the admitted market side reveals which universal claim is at issue. A pathwise claim concerns individual price paths, rather than averages over a probability law.

A strategy is *universal* here only if, after discounting by a specified numeraire, it produces strictly positive terminal gain from zero initial capital on every admitted trajectory at a specified finite horizon. The numeraire is a positive-valued reference asset, here a bank account; discounting expresses prices and gains in units of that asset. The horizon is the final evaluation date, and a trajectory is a price path. This claim differs from nonnegative gain everywhere with positive gain somewhere, eventual profit at a path-dependent date, expected profit (a probability-weighted average), and small regret to a benchmark. A benchmark is a specified reference investment or strategy; regret measures a strategy's performance shortfall relative to it. Zero cost means zero initial investment, self-financing means no later external cash injections, and admissibility imposes restrictions on allowable trading, such as a lower bound on wealth. These conditions distinguish trading gain from positive wealth obtained by investing positive capital. The order of the quantifiers, which specify "there exists" and "for every," is

$$\exists\, \sigma\ \forall\, \mathcal{E}: \qquad \sigma \text{ wins against } \mathcal{E}. \tag{1}$$

Here σ denotes a trading strategy and $\mathcal{E}$ an admitted market; $\exists$ means "there exists" and $\forall$ means "for every." "Wins" means earning strictly positive discounted gain from zero initial capital at the specified finite horizon. The order requires one fixed strategy to succeed against every admitted market. By contrast, diagonalization—constructing a counterexample tailored to the proposed strategy—establishes

$$\forall\, \sigma\ \exists\, \mathcal{E}_\sigma: \qquad \sigma \text{ does not win against } \mathcal{E}_\sigma. \tag{2}$$

The subscript in $\mathcal{E}_\sigma$ indicates that the countermarket may depend on the chosen strategy; there need not be one market defeating all strategies.

A computable rule is one executable by an algorithm; "effective" and "recursive" have this same computational meaning here. A total rule returns an answer on every allowed input, and a deterministic rule makes no random choices. For a fixed total deterministic computable trader, its code can be embedded in a total program that recursively chooses the next bounded proportional price move opposite to each nonzero position and a nonzero move during inactivity. The resulting countermarket is one fixed computable path; any finite prefix can be generated offline. A trader that never takes a position earns zero, which already refutes a guarantee of strict profit. This is the trading counterpart of the next-bit construction $x_0 = 1 - F(\varnothing)$ and $x_t = 1 - F(x_0, \ldots, x_{t-1})$ for $t \geq 1$: it defeats the fixed proposed rule, not every rule on one exceptional sequence. Here a bit is either zero or one, F predicts the next bit from the preceding bits, $\varnothing$ denotes the empty history, and x_t is the bit chosen opposite to that prediction at date t. A prefix is an initial finite segment of a sequence; generating it offline means computing it without interacting with a live market.

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The opponent-specific character of the construction also permits a converse. Given a total computable price program with nonzero changes, a trader can compute the next change and choose its profitable sign. Section IX establishes this countertrader result and distinguishes separate counterconstructions from mutually consistent responses in a single interaction. Code access does not give both sides the ability to respond last. Section VII connects bounded-credit trading to betting martingales and distinguishes computable from lower-semicomputable capital processes.

Three complementary limits use different market information. Gold's learner sees an accumulating history and must eventually stabilize on a correct program; no effective learner identifies every binary total recursive time function (a program assigning a bit to every date), although restricted classes such as the primitive-recursive time functions are identifiable in the limit [3, 4]. Primitive-recursive functions are built from elementary functions using composition and recursion whose number of stages is set by the input. Identification in the limit means eventually keeping one correct rule, without necessarily knowing when that point has been reached. A source-code procedure that settled every future event of Turing-universal generators, which can simulate any program, would instead decide the Halting Problem: whether an arbitrary program eventually stops. Busy-Beaver growth, the growth of the longest halting runtime among programs up to a given length, excludes a computable worst-case waiting horizon. Finally, relative to a specified computable probability law, a Martin-Löf-random record escapes every effective null test: an algorithmically specified test whose full failure set has probability zero. Such records have measure one, meaning probability one, also called "almost surely." Finite samples cannot certify this infinite-sequence property.

Repeatable practical success must therefore identify structure absent from the unrestricted market class: persistence (continuation of past behavior), mean reversion (movement back toward a reference level), a risk premium (expected compensation for bearing risk), a benchmark, lawful information, or another distributional asymmetry (an imbalance in possible outcomes or their probabilities). An edge is an expected performance advantage under specified market conditions. Large, crowded, or public strategies may also change the distribution through price impact, meaning price changes caused by their own orders, and adaptation by other participants [5]. This falsifiable restriction, one that evidence could contradict, is the operational meaning of "insight" used below.

The explicit diagonal countermarket is the constructive core of the paper. Table I organizes the complementary limits into a taxonomy that keeps distinct several statements commonly summarized as "no universal strategy." The construction and its proofs are expository reformulations rather than a claim of a newly discovered impossibility theorem. The analysis connects this quantifier map to the literature on unpredictable prices [6–8], Gold learning, computability and randomness, time-reversal and permutation stress testing (reversing or otherwise reordering a record), and the conditions under which strategies may work "for all practical purposes" (FAPP, following Bell [9]).

II. A COMPUTABLE COUNTERMARKET FOR EVERY TRADER

The trader is a program M_σ mapping each observed finite price history to a position; the market side supplies the price history.

Throughout the discrete constructions, dates $t = 0, 1, 2, \ldots$ count trading periods, and T denotes a finite final date. The quantity $P_t = S_t/B_t$ is the discounted price of one asset, where S_t is its price in a fixed currency and $B_t > 0$ is the value of one unit of the chosen risk-free bank account. For example, if both initially equal 100, and later $S_1 = 121$ and $B_1 = 110$, the discounted price rises from 1 to 1.1. This measures performance relative to the bank account, not inflation. The position w_t is a number of shares chosen before P_{t+1} is observed. A positive position holds shares, a negative position sells borrowed shares short, and zero means no asset exposure. A short position profits from a price fall and loses from a rise. Trading is self-financing: changes in holdings are funded through the existing share and bank balances, with no external cash injections. Let V_t be total portfolio wealth at date t, measured in bank-account units. More explicitly, holding w_t shares and the residual $V_t - w_t P_t$ in the discounted bank account gives

$$V_{t+1} = V_t + w_t(P_{t+1} - P_t).$$

We omit transaction costs, borrowing limits, leverage caps (limits on exposure relative to capital), and solvency constraints (requirements to remain able to meet obligations). The counterexample therefore also applies to any subclass of these strategies satisfying additional restrictions.

Let $M_\sigma \colon \mathbb{Q}_{>0}^{<\omega} \to \mathbb{Q}$ be a total computable function on finite rational histories, encoded in the usual effective way, with $w_t = M_\sigma(P_0, \ldots, P_t)$. Here $\mathbb{Q}$ is the set of rational numbers, $\mathbb{Q}_{>0}$ its positive members, and the superscript $< \omega$ denotes all finite sequences; the arrow specifies the function's input and output sets. Totality requires an action in finite computation time on every input; it supplies no prescribed running-time bound or guarantee of meeting a trading deadline. Here the program outputs an exact rational representation, not merely a procedure approximating a real number. Such outputs model finite-precision orders and make comparison with zero decidable. Let $\mathcal{P}^{+}_{\mathrm{comp}}$ denote the positive rational-valued computable price paths.

TABLE I. Distinct claims organized by the trader–market configuration. They are complementary, not interchangeable. Here P_σ is the price path constructed for strategy σ; a black box reveals outputs but not its internal program, and a promise restricts the inputs on which an algorithm must work.

Mechanism	**Market side**	**Formal scope**	**Conclusion**
Diagonal countermarket	Fixed total program constructed from the fixed trader's code	Pathwise worst case: $\forall\sigma\,\exists P_\sigma$	No total deterministic computable trader with exact rational outputs wins on every positive rational-valued computable price path
Known-program countertrader	Supplied total computable price program with nonzero changes	Opponent-specific construction; every positive finite horizon	Each such path admits a profitable computable trader; the two perfect current responses cannot hold together
Gold identification in the limit	Accumulating computable history, or interaction with a black box	Success is eventual stabilization on a correct program, without a known stopping date	No effective learner identifies every computable binary time function; restricted classes can be identified
Halting reduction and Busy Beaver	Total program or generator, under an explicit promise	Individual encoded computation over an unbounded horizon	No general computable profit or market-event decider, or computable description-size waiting horizon
Algorithmic randomness	Passive binary record under the computable fair-coin reference law	Each Martin-Löf-random path; such paths have measure one	Independent fair trials yield Martin-Löf-random records almost surely; finite tests cannot certify randomness
Betting martingales	Binary fixed-magnitude paths; bounded-credit trading	Unbounded capital along an infinite record	Computable and lower-semicomputable capital characterize different randomness notions; finite-horizon success has a probability bound

Theorem 1 (Computable Diagonal Countermarket)**.** *Fix a rational initial price $P_0 > 0$ and a rational proportional price-change magnitude $0 < \epsilon < 1$. For every total deterministic computable strategy M_σ that returns exact rational positions, there exists a fixed path $P^{M_\sigma} \in \mathcal{P}^+_{\mathrm{comp}}$, starting at P_0, with $P_{t+1}/P_t \in \{1-\epsilon, 1+\epsilon\}$ at every date, on which its self-financing gain*

$$G_T = \sum_{t=0}^{T-1} w_t(P_{t+1} - P_t)$$

is non-positive at every finite horizon T. It is strictly negative after the strategy has taken a nonzero position at least once.

The superscript in P^{M_σ} labels dependence on the trader's program; it is not a power. The gain G_T sums the position times the price change over the first T periods and equals final discounted wealth when $V_0 = 0$. The proportional return is the price change divided by the preceding price; its magnitude here is always ϵ.

Proof. Choose rational $P_0 > 0$ and rational $0 < \epsilon < 1$. Recursively compute $w_t = M_\sigma(P_0, \ldots, P_t)$ and set

$$P_{t+1} = \begin{cases} P_t(1-\epsilon), & w_t > 0, \\ P_t(1+\epsilon), & w_t < 0, \\ P_t(1+\epsilon(-1)^t), & w_t = 0. \end{cases} \tag{3}$$

Because M_σ is total and rational-valued, this recursion is a total program for one fixed infinite price path. Its prices remain positive and computable, and every return has magnitude ϵ. In each period, $w_t(P_{t+1} - P_t) \le 0$, with strict inequality when $w_t \neq 0$. Summing proves the claim at every finite horizon. □

In the last case of Eq. (3), the alternating market move is a convenient but arbitrary choice. Since $w_t = 0$, any market move yields zero gain in that period. Within the theorem's fixed-magnitude class, any computable choice between returns $+\epsilon$ and $-\epsilon$ is admissible, allowing a wide variety of market-move sequences in place of the alternating convention.

For a concrete example, fix the rule that holds one share for the first period, shorts one share for the second, and then holds no shares. With $P_0 = 100$ and $\epsilon = 1/10$, Eq. (3) produces

$$(w_0, w_1, w_2) = (1, -1, 0),$$
$$(P_0, P_1, P_2, P_3) = (100, 90, 99, 108.9).$$

The successive discounted gains are $-10, -9, 0$, so $G_3 = -19$, all in bank-account units. The prices are constructed after fixing the rule; they are illustrative, not market data.

Remark 1 (Logical and Economic Scope of the Countermarket). Theorem 1 uses the trader's *code*, not a physical market's observation of a deployed order. Once M_σ is fixed, the countermarket program is fixed and can generate any desired finite prefix offline. It is therefore an admissible deterministic computable counterexample within the stated model, although not necessarily an efficient one. Computability and economic rationality are separate assumptions. If an artificial opponent is assigned the objective of preventing this trader's strict gain, the rule serves that objective; it is not thereby an economic equilibrium or an arbitrage-free price process. Another trader may profit on the resulting path and has, in turn, its own computable countermarket. Section IX makes this converse and its limits explicit.

Code access is an information assumption of the construction. Without it, a market required to forecast the next buy-or-sell action from past actions faces the direct next-bit obstruction of Section III: every total computable predictor has a computable time-dependent trading rule against which each forecast is wrong. Identifying the trader's generating rule is a different task. Since time-dependent buy-or-sell rules can generate every computable binary sequence, Gold's identification limit also applies to that task [3, 4]. Neither statement establishes the failure of every antagonistic action. Unbounded questions about a trader, such as whether it will ever take a nonzero position, can also encode the Halting Problem and remain undecidable even when its code is supplied (Section IV). If the market instead observes the current position before setting the next price, it can oppose that position directly without knowing or inferring the strategy.

This is a direct strategy-relative diagonal construction. The proposed universal trader is used to compute an object outside its success set: long is paired with down, short with up, and neutrality with an alternating nonzero return. Just as the complement sequence defeats a purported predictor of every computable sequence, P^{M_σ} defeats the purported trader on every horizon. Enumeration of all programs is unnecessary because the universal claim itself supplies the program to be diagonalized against. The quantifiers are $\forall M_\sigma \, \exists P^{M_\sigma}$. A constant discounted price path already gives every self-financing rule zero trading gain and refutes strict profit on a class containing that path. The stronger content of Theorem 1 is that every nonzero position of the selected trader incurs a strict loss even on the class with fixed return magnitude ϵ, which contains no flat path. It does not supply a path on which every strategy strictly loses: permanently long and permanently short positions cannot both lose on the same one-step price move.

Remark 2 (Exact Outputs and Computable-Real Positions). Theorem 1 assumes that each position is returned as an exact rational number, whose sign is decidable. An alternative representation provides, on input a rational price history and precision k, a rational approximation to the position with error at most 2^{-k}. The approximation algorithm must be total and uniform in both inputs. Strict positivity and strict negativity are then semidecidable, but deciding between them and zero need not be possible. The distinction concerns the effective representation: even positions whose values are all dyadic rationals can exhibit this problem. A generic sign search may wait forever at an exactly zero position, so Eq. (3) need not define a total program in this representation. Finite-precision executed orders fit the exact-output model; the approximation-based model is a mathematical idealization. Empty price histories may be assigned position zero throughout.

Proposition 2 (Failure of Per-Period Opposition)**.** *Fix P_0 and ϵ as in Theorem 1, retaining the same price-path class but allowing approximation-based outputs instead of exact rational outputs. There is a trader with uniformly computable-real positions, depending only on the date, such that every computable path in the fixed-magnitude class gives it a strictly positive gain in at least one period.*

Proof. Enumerate the partial computable functions $\varphi_e \colon \mathbb{N} \rightharpoonup \{-1, +1\}$. If $\varphi_e(e)$ halts after $n \geq 1$ steps with output b, set $w_e = b2^{-n}$; otherwise set $w_e = 0$. Simulating for k steps and returning the exact value if it has appeared, and zero otherwise, approximates w_e within 2^{-k}, uniformly in e. Use the history length minus one to obtain the date. For any computable path in the stated class, $d_t = \operatorname{sgn}(P_{t+1} - P_t)$ is a total computable sign sequence. Choose an index e with $\varphi_e = d$. Then $w_e \neq 0$ has sign d_e, so $w_e(P_{e+1} - P_e) > 0$. □

The two results concern different representations of traders on the same class of paths. The positions just constructed are dyadic rationals, but no uniform algorithm returns their exact rational values or decides their signs including zero. Code access therefore does not make Eq. (3) executable for this trader. The proof covers every computable path, including any produced from the trader's code; it makes no independence assumption about the path. A positive gain in one period also need not make cumulative gain positive.

Proposition 3 (Cumulative Nonpositivity Survives)**.** *For every trader whose positions on rational histories are uniformly computable reals, there exists a computable path in the same fixed-magnitude class with $G_T \leq 0$ at every finite horizon.*

Proof. Consider the computable all-up path $P_t = P_0(1 + \epsilon)^t$. If every position along it is zero, it is already the required path. Otherwise let s be its first nonzero-position date. Retain its history through P_s, then choose the return opposite to that position. This gives $G_T = 0$ for $T \leq s$ and $G_{s+1} < 0$. The finite prefix and this direction can be hard-coded.

Thereafter, given $G_t < 0$, the two possible next cumulative gains are

$$A_+ = G_t + \epsilon P_t w_t, \qquad A_- = G_t - \epsilon P_t w_t.$$

Both are computable reals and their average is $G_t < 0$; hence at least one is strictly negative. Run their strict-negativity tests in parallel and choose the direction whose test first certifies negativity. Each step terminates and preserves $G_{t+1} < 0$, generating a computable infinite price path. The initial case distinction and finite-prefix selection establish existence, without providing a uniform procedure from trader code to countermarket code. □

Thus approximated positions invalidate the guarantee that *every* active position loses, but not the existence of a computable countermarket preventing positive cumulative gain at every horizon. Given the trader's approximation program, a Halting-Problem oracle can decide each sign, including zero, and recover the per-period construction relative to $\emptyset'$.

III. LEARNING A MARKET RULE FROM HISTORY: GOLD'S LIMIT

A trader normally receives neither the market generator's source code nor a proof of its law. The trader sees an expanding record $x_0, x_1, \ldots$ and must infer from each finite prefix how it will continue. Gold's function-identification model formalizes exactly this information flow. At time n, an effective learner L, itself an algorithm, maps $(x_0, \ldots, x_n)$ to the index, or numbered program description, of a Turing machine, an abstract model of computation. It *identifies x in the limit* if, from some finite date onward, every guess is the same index and that machine computes the whole sequence. The learner need not know when its last change of mind has occurred [3, 4].

Let REC_2 denote the total computable binary sequences. Gold's negative theorem for binary-valued time functions has the trader–market quantifier order, with L ranging over effective learners,

$$\begin{aligned} &\forall L\ \exists x \in \mathrm{REC}_2 : \\ &\quad L(x_0, \ldots, x_n) \text{ does not stabilize} \\ &\quad \text{on a program for } x. \end{aligned} \tag{4}$$

The adverse history depends on the learner. This is closely analogous to $\forall M_\sigma\ \exists P^{M_\sigma}$, although Gold's failure criterion is failure to converge to a fixed correct program name, not financial loss.

The corresponding claim about literal next-move forecasting is sharper and has the short direct diagonal proof already used above. If a total computable predictor F outputs one bit from every finite binary history, the recursive definition

$$x_0 = 1 - F(\varnothing), \qquad x_t = 1 - F(x_0, \ldots, x_{t-1}) \quad (t \geq 1)$$

produces a computable market record on which F is wrong at every date. Thus the proposed statement—every total next-move predictor has a computable countermarket—is correct, but it is this direct prediction diagonalization, not the literal statement of Gold's identification theorem. Nonconvergence of program names alone does not count the learner's next-step errors.

The opposite-bit construction also appears explicitly in Legg's analysis of universal prediction [10]. The qualifier *computable* is essential. Solomonoff induction combines program-based predictive models, giving shorter descriptions greater weight. It defines an uncomputable universal predictive ideal; for computable probabilistic environments its conditional predictions converge, with quantitative error bounds developed by Hutter [11–14]. It therefore evades, rather than contradicts, the diagonal result: it is not a total executable trading rule, and predictive convergence is not Gold's requirement of stabilizing on one exact generating program. For probability-valued forecasts, V'yugin gives a further, distinct impossibility: a randomized construction can make any partial computable forecaster fail calibration with probability arbitrarily close to one [15]. A partial algorithm may fail to return an answer on some inputs; calibration requires forecast probabilities to agree with observed frequencies on the corresponding forecast groups. Calibration, next-bit error, and identification in the limit are three different criteria.

Gold's results also show why the restriction on the market side matters. Every uniformly effective indexed family of total computable functions—one algorithm computes any member from its index and input—is identifiable in the limit: a learner selects the least index consistent with the observed prefix, and eventually eliminates every smaller incorrect index. In particular, primitive-recursive time functions form such a family. In Gold's interactive black-box model, where the eventual hypothesis must reproduce future input–output behavior, finite-state boxes, with only finitely many internal states, are weakly identifiable in the limit, whereas the class of primitive-recursive black boxes is not [3]. These are different object classes, not contradictory claims. Gold's 1967 language model further makes explicit that learnability depends on the admitted class, the presentation of the evidence, and the allowed names [4]. For trading, a forecast can therefore be justified only relative to a specified restriction on possible markets.

IV. UNIVERSAL COMPUTATION AND THE HALTING BARRIER

Gold's learner works from observations and may revise its hypothesis indefinitely. The next results ask a separate, stronger operational question: can one terminating program inspect arbitrary encoded traders or market generators and correctly decide their unbounded future behavior? The preceding diagonal construction already

disproves a trader that wins on every computable path and does not invoke the Halting Problem. The source-code event-certification question does. A problem is undecidable if no algorithm can always terminate with a correct yes-or-no answer.

Theorem 4 (Undecidability of Eventual Profit)**.** *Fix the positive computable price path $P_t = t+1$, $t = 0, 1, \ldots$. There is no algorithm which, under the promise that its input code computes a total trading program M_σ, correctly decides for every such code whether*

$$\exists T \geq 1 : \qquad G_T(M_\sigma; P) = \sum_{t=0}^{T-1} w_t(P_{t+1} - P_t) > 0.$$

Proof. Assume such a profit certifier exists. Given an arbitrary Turing machine A and input x, construct a trading program $M_{A,x}$ as follows. At date t, simulate $A(x)$ for $t+1$ steps. If it has halted within those steps, set $w_t = 1$; otherwise set $w_t = 0$. Every date requires only a finite simulation, so $M_{A,x}$ is total. Along $P_t = t+1$, every long position earns one unit in the next period. Hence $G_T > 0$ for some T if and only if $A(x)$ eventually halts. A certifier for eventual trading profit would therefore decide the Halting Problem, which is impossible [16]. □

This reduction, a conversion of one decision problem into another that preserves its answer, can also be placed directly on the market side, which is the form most closely connected to source-code event certification.

Theorem 5 (Undecidability of a Market Event)**.** *There is no algorithm which, under the promise that its input code computes a total positive-price generator G, correctly decides for every such code whether its generated path P^G ever satisfies $P^G_t > P^G_0$.*

Proof. Given a Turing machine A and input x, define the total generator by $P_0^{A,x} = 1$ and, for $t \geq 1$,

$$P_t^{A,x} = 1 + \mathbf{1}\{A(x) \text{ halts within } t \text{ steps}\}, \qquad t \geq 1. \quad (5)$$

Here $\mathbf{1}\{\cdot\}$ is the indicator: one when its condition holds and zero otherwise. Each quoted price requires only a finite t-step simulation, so the generator is total and computable. Its initial price is one, and it ever quotes a higher price if and only if $A(x)$ halts. A universal decider for the stated market event would therefore decide the Halting Problem. □

Theorem 5 makes "universal computation" precise: the admissible generators can emulate an arbitrary machine and expose its outcome through a price event. No algorithm can then certify every future event of every such market. Both reductions concern promise domains: the putative decider must halt and be correct on every promised total input. Positive instances are semidecidable: simulation confirms them when the event occurs, but may run forever when it does not. There is no terminating general procedure for certifying its perpetual absence.

This is not Gold's theorem. The reduction receives source code and must halt with a yes-or-no answer; Gold's learner receives accumulating data and need only stabilize eventually. Indeed, the paths in Eq. (5) are identifiable in the limit: guess the flat path, then switch permanently if the price becomes two. No finite date, however, certifies that the initial guess is final. Thus halting excludes uniform certification, whereas Gold excludes identification in the limit for the class of all binary total recursive time functions. Restricted model classes may still be learnable [17–21].

Rice's theorem gives a related general limit: any property that depends only on the function computed, holds for some partial computable functions, and fails for others is undecidable for arbitrary programs [22]. Totality itself is undecidable in general [16]. This does not prevent evaluating a fixed total strategy on a completed finite computable path by simulation. The explicit reductions above establish the restricted claim needed here: future-event and eventual-profit questions remain undecidable even when the input program is promised to be total.

a. Busy-Beaver horizons: finite glimpses and non-computable waiting Fix a step-counted universal machine V with an effective compiler that adds only fixed overhead for a given program schema, or template. The compiler translates descriptions into code for V; fixed overhead is a constant number of extra code bits. Define the runtime Busy Beaver function by

$$\mathrm{BB}(n) = \max\{\mathrm{time}_V(p) : |p| \leq n \text{ and } V(p) \text{ halts}\}, \quad (6)$$

with value zero if the set is empty. Here p is a program, $|p|$ its length in bits, and $\mathrm{time}_V(p)$ its number of execution steps on V. The input n is a program-description budget, not a date; $\mathrm{BB}(n)$ is a runtime, which becomes a possible trading date when the computation is embedded in a market generator. A small increase in rule length may therefore require waiting beyond every computably prescribed scale.

Each $\mathrm{BB}(n)$ is finite, and selected low values can sometimes be proved by exhaustive simulation plus proofs that the remaining programs do not halt. This is not uniform "computability for small arguments": any finite table can be hard-coded, and its apparent pattern licenses no extrapolation. If a total computable h upper-bounded BB, simulating every program of length at most n for $h(n)$ steps would decide halting. Eventual domination uses a further program-schema argument. For each n, a program of $O(\log n)$ bits—at most a constant times $\log n$ for sufficiently large n—can encode n, compute $h(n)$, execute more than $h(n)$ computation steps solely to delay termination, and halt. Here and below, log denotes the natural logarithm. For all sufficiently large n, its compiled V-code has length at most n, so $\mathrm{BB}(n) > h(n)$. Thus, up to the fixed machine convention, BB eventually exceeds every total computable function [23, 24].

Calude, Dinneen, and Shu illustrate the finite/infinite divide. For one explicit universal machine U with prefix-free program codes, meaning no valid code is a proper

prefix of another, computation and mathematical proof settled halting through 84-bit programs and certified 64 initial bits of its halting probability Ω_U. This is the probability that U halts when its program bits are supplied by independent fair coin tosses. They did not compute Busy Beaver values, and the work needed to recover the relevant halting programs from longer supplied prefixes of Ω_U outruns every computable bound. Their result is a formidable finite glimpse, not a scalable computation of an uncomputable object [25].

Applied to Eq. (5), a short generator can quote $P_t = 1$ for a Busy-Beaver-scale interval and then quote $P_t = 2$. Across encoded programs of length at most n, the last halt can occur at $\mathrm{BB}(n)$, up to fixed wrapper overhead. Knowing that horizon would make continued silence a certificate of no later move. Equivalently, with TotComp denoting total computable size-dependent schedules,

$$\forall h \in \mathrm{TotComp}\ \exists p: \quad \begin{array}{l} V(p) \text{ halts}, \\ \mathrm{time}_V(p) > h(|p|). \end{array} \tag{7}$$

If c is the wrapper's fixed description-length overhead, it is absorbed by the computable envelope $H(n) = \max_{k \le n+c} h(k)$. Every quote at a specified finite date remains computable by bounded simulation; the generator neither computes Busy Beaver nor selects its maximizer uniformly. Noncomputability lies in certification across the family. Each realized path is deterministic and computable, hence not Martin-Löf random under fair-coin coding.

This barrier is stronger than sensitive dependence on initial conditions or visual complexity alone in a precise decision-theoretic sense. Repeatedly applying finitely many distance-shrinking maps, or a chaotic map in which nearby initial states can separate strongly, can generate intricate output from a short computable rule. Turing universality additionally creates questions about the unbounded future for which no terminating general algorithm exists. The categories can overlap when a dynamical system performs universal computation. This is a worst-case claim across generator descriptions, not evidence that typical financial markets exhibit Busy-Beaver delays [21, 26, 27].

These obstructions concern unbounded horizons. At any specified finite horizon, total programs can still be run and their realized prices and gains computed.

V. WORST-CASE VALUE AND PRIVATE RANDOMIZATION

Returning to the direct countermarket, its finite-horizon value can be stated explicitly, and the role of a trader's hidden randomization can be separated from deterministic pathwise opposition. The maximin value is the best worst-case lower bound on gain: the inner infimum ranges over markets, and the outer supremum ranges over strategies.

Corollary 6 (Maximin Value)**.** *At every finite horizon T, if the neutral strategy (always holding zero shares) is allowed, then*

$$\sup_{\text{total computable } M_\sigma} \inf_{P \in \mathcal{P}^+_{\mathrm{comp}}} G_T(M_\sigma, P) = 0.$$

Proof. The neutral strategy guarantees zero, while Theorem 1 gives every strategy a computable countermarket with gain at most zero. □

a. Two sources of randomness Randomness can enter on either side: *private trader randomization* randomly selects positions, while *market randomness* generates price movements. They can occur separately or together and need not have the same effect on profits. Private randomization need not destroy an existing edge, and random market movements can have positive expected discounted returns. The result below uses the stronger assumption that the next discounted price change has conditional mean zero, even given the trader's private seed, the random input used to choose positions. Thus the expected next change, given all available information, is zero. Integrability means finite expected absolute value; under this condition the gain has expectation zero at each fixed horizon, not convergence of realized profits to zero in the long run.

Proposition 7 (Private Randomization Does Not Create a Guarantee)**.** *Fix a finite horizon T, a rational $P_0 > 0$, and a rational $\epsilon \in (0, 1)$. There is a single passive computable market law, independent of the trader, under which every predictable randomized strategy, choosing positions using only information already available, with integrable gains has expected gain zero. Specifically, let*

$$P_{t+1} = P_t(1 + \epsilon \xi_{t+1}), \qquad t = 0, \ldots, T-1,$$

where $(\xi_t)_{t \ge 1}$ are independent fair signs, each equal to $+1$ or -1 with probability one half, also independent of the trader's private random seed. If w_t is measurable with respect to the information available when the position is chosen and $w_t(P_{t+1} - P_t)$ is integrable, then

$$G_n = \sum_{t=0}^{n-1} w_t(P_{t+1} - P_t), \qquad n = 0, \ldots, T,$$

is a martingale: its expected next value, conditional on current information, equals its current value. In particular, $\mathbb{E}[G_T] = 0$, where $\mathbb{E}$ denotes expectation.

Proof. Let R denote the trader's private seed and let $\mathcal{G}_t$ be the sigma-algebra generated by $P_0, \ldots, P_t, R$: the collection of events determined by the observed prices and the private seed. Allowing $\mathcal{G}_t$ to contain all of R only strengthens the claim. Predictability makes w_t $\mathcal{G}_t$-measurable, meaning determined by this information, while ξ_{t+1} is independent of $\mathcal{G}_t$ and has mean zero. Hence

$$\mathbb{E}[w_t(P_{t+1} - P_t) \mid \mathcal{G}_t] = \epsilon P_t w_t \, \mathbb{E}[\xi_{t+1} \mid \mathcal{G}_t] = 0.$$

Here $\mathbb{E}[\cdot \mid \mathcal{G}_t]$ is expectation conditional on that information. The integrability assumption therefore makes $(G_n)_{n=0}^T$ a martingale, and $\mathbb{E}[G_T] = \mathbb{E}[G_0] = 0$. □

Private randomization prevents a deterministic countermarket based only on the public history from mechanically opposing the unseen current draw, but it does not yield strictly positive expected gain in every environment. More precisely, under the proposition's integrability assumptions it excludes $\forall P : \mathbb{P}_R(G_T(P, R) > 0) = 1$ at the fixed horizon T, where $\mathbb{P}_R$ takes probability over the trader's seed for a fixed path. Each of the finitely many fair-sign prefixes has positive probability and a computable infinite continuation; such a guarantee would imply $G_T > 0$ almost surely jointly, hence $\mathbb{E}[G_T] > 0$, a contradiction. It does not exclude a high win probability, profit under another law, or benchmark-relative success.

b. Optional online-game reading The recursion in Theorem 1 also defines a finite perfect-information game, in which both players observe every previous move: the trader announces w_t, the environment chooses the next return, and Corollary 6 gives the trader-first maximin value zero. This reading is optional because the fixed deterministic trader's countermarket program already generates the same path offline. Borel determinacy concerns infinite games whose winning sets are generated from finite-history events by complements and countable unions (Borel sets); it does not by itself identify a winner here [28]. Hidden private randomization is a different information structure: Proposition 7 proves zero expected gain under one passive computable fair-sign law, not a value for a game with randomized strategies without specified strategy spaces. Section IX specifies a simultaneous game and distinguishes its mixed-strategy equilibrium from this martingale statement.

VI. A PASSIVE MARKET: MARTIN-LÖF RANDOMNESS

a. When does process randomness yield a random record? Process randomness describes how individual bits are generated and depend on their past. Martin-Löf (ML) randomness is a property of the resulting *infinite sequence* relative to a specified probability law. *Independent fair bits produce a fair-coin ML-random sequence with probability one.* Equivalently, the sufficient process condition is

$$\mathbb{P}(X_{n+1} = 1 \mid X_1, \ldots, X_n) = \tfrac{1}{2} \quad \text{a.s. for every } n \geq 0, \tag{8}$$

Here X_n is the random bit at date n, $\mathbb{P}$ denotes probability, and "a.s." means almost surely, or with probability one. The condition includes the first bit at $n = 0$.

Independence alone is insufficient for *fair-coin* randomness: independent bits with a fixed probability $p \neq 1/2$ of being one almost surely have limiting frequency p. Fair one-bit probabilities alone are also insufficient: repeating one fair bit forever gives fair marginals—each bit separately is equally likely to be zero or one—but a constant, predictable record. Independence is not necessary for ML randomness relative to the process's *own* computable law, which may include bias or dependence.

For FAPP use, the conclusion is therefore conditional on a justified source model, for example independent fair trials. Tests for bias and serial dependence (statistical dependence between different dates) can challenge that model over observed scales; passing them cannot certify independence indefinitely or the ML randomness of an infinite output. An almost-sure conclusion allows exceptional records of probability zero. A fixed deterministic pseudorandom generator with a finite seed cannot produce a fair-coin ML-random infinite sequence, however well its finite outputs test.

Here is the precise classical result. A computable law μ has cylinder probabilities $\mu([s])$ uniformly approximable to any requested accuracy, where s is a finite bit string and its cylinder $[s]$ comprises all infinite sequences beginning with s. An ML test is a uniformly computably enumerable sequence of open sets (U_k) with $\mu(U_k) \leq 2^{-k}$: each set is a union of cylinders, and one algorithm lists their defining prefixes given k. A record is μ-ML-random if it avoids $\bigcap_k U_k$, the set common to every level, for every such test. Write MLR_μ for the set of these random records.

Proposition 8 (Classical Measure-One Result, Martin-Löf)**.** *If a binary process has computable joint law μ, then*

$$\mathbb{P}\big((X_1, X_2, \ldots) \in \mathrm{MLR}_\mu\big) = 1. \tag{9}$$

This includes independent fair trials, independent bits with a fixed computable probability of one (a Bernoulli bias), and computable dependent laws [29, Sec. IV].

Proof. Each test's failure set has measure zero. There are countably many effective tests, so their union still has measure zero. □

The law must be specified: randomness relative to an atomic law, which gives positive probability to individual records, need not mean fresh uncertainty at each trial. For example, either constant record is random for the law that repeats one fair bit forever.

b. Borel normality is weaker A sequence is Borel normal in base b when every word, or contiguous block, of k digits has limiting frequency b^{-k}, for every $k \geq 1$; the digit alphabet is $0, \ldots, b-1$. ML randomness for uniform base-b digits implies normality, but the converse fails. Champernowne's decimal $0.123456789101112\ldots$, formed by concatenating the positive integers, is normal in base ten yet computable, hence not ML-random for the uniform decimal law [30]. Normality is difficult to establish for a given infinite record: one must control all block lengths in the infinite limit. No finite frequency check proves it, since any finite prefix admits both normal and nonnormal continuations. A proof from a construction,

as for Champernowne, can establish normality; extensive sampling alone cannot.

VII. TRADING, MARTINGALES, AND ALGORITHMIC RANDOMNESS

Effective randomness can also be expressed through the inability of specified betting strategies to accumulate unbounded capital. The computational class of the strategy matters: computable randomness and Martin-Löf randomness use different classes [31–33]. This betting viewpoint complements Levin's characterization using the monotone description complexity of initial segments [34].

Fix rational $P_0 > 0$, rational $0 < \epsilon < 1$, and initial credit $C > 0$. For $x = (x_1, x_2, \ldots) \in \{0,1\}^{\mathbb{N}}$, set

$$P_t(x) = P_0 \prod_{i=1}^{t} (1 + \epsilon(2x_i - 1)).$$

Each price history in this class determines its bit string. A trader is *admissible with credit* C if its gain satisfies $G_n \geq -C$ at every date on every such history. Thus $C + G_n$ is nonnegative wealth. A nonnegative function d on finite binary strings is a *martingale* if

$$d(s) = \tfrac{1}{2}\big(d(s0) + d(s1)\big).$$

It *succeeds* on x when $\limsup_n d(x \upharpoonright n) = \infty$, where $x \upharpoonright n$ is the first n bits.

Proposition 9 (Traders and Martingales)**.** *Total computable traders returning exact rational positions and admissible with rational credit C correspond exactly, on this price class, to nonnegative martingales with exactly computed rational values and $d(\emptyset) = C$. The correspondence and its inverse are*

$$d(s) = C + G_{|s|}(s), \qquad w_{|s|}(s) = \frac{d(s1) - d(s0)}{2\epsilon P_{|s|}(s)}.$$

Proof. The two next gains are $\pm w_{|s|}(s)\epsilon P_{|s|}(s)$, so their average is zero. Conversely, the displayed position gives gain $d(sb) - d(s)$ on extension by b; these increments telescope. Nonnegativity is precisely admissibility. All quantities are rational and computable. Set the trader's position to zero on price histories outside the specified class, including the empty history, to obtain a total rule. □

Theorem 10 (Effective Betting Characterizations)**.** *A binary record x is computably random if and only if every total computable trader returning exact rational positions and admissible with some rational credit has bounded gain along $P(x)$. It is Martin-Löf random if and only if no nonnegative left-computably-enumerable martingale succeeds on it.*

Here *left-computably-enumerable*, or *lower semicomputable*, means that the values have uniformly computable increasing rational approximations, without a required computable error bound. The standard betting characterizations give the theorem [31–33]. Restricting computable martingales to exact rational outputs does not change computable randomness. Given a computable-real martingale d, choose rational initial capital greater than $d(\emptyset) + 1$. Approximate each capital wager $(d(s1) - d(s0))/2$ by a rational within $2^{-|s|-2}$, and update the new capital by adding or subtracting that rational. Along every path the accumulated error is less than $1/2$. The resulting rational martingale stays nonnegative and differs from d by a bounded amount, so preserves success.

The second characterization must not be read as an equivalence with our executable rational traders. A left-computably-enumerable martingale may have noncomputable initial capital. Indeed, if its initial capital is computable, its averaging identities and lower approximations give upper approximations too, making all its values computable reals. Furthermore, the difference $d(s1) - d(s0)$ in the position formula need not be lower semicomputable or rational. Lower approximations to wealth therefore do not by themselves produce computable orders.

There is nevertheless one left-computably-enumerable martingale succeeding on every non-Martin-Löf-random record. To see this, take a universal Martin-Löf test $(U_n)_{n\geq 1}$, with fair-coin measure $\mu(U_n) \leq 2^{-n}$, and put

$$D(s) = \sum_{n \geq 1} \frac{\mu(U_n \cap [s])}{\mu([s])}.$$

Here $[s]$ is the set of infinite extensions of s. The sum is finite at every string, is lower semicomputable, and satisfies the martingale identity, with $D(\emptyset) \leq 1$. If $x \in \bigcap_n U_n$, each of the first arbitrarily many summands eventually equals one along x, proving success. Universality here concerns successful records; it does not assert multiplicative domination of every left-computably-enumerable martingale. In particular, it is not obtained by effectively listing all such martingales and taking their weighted sum [32, 33].

For any specified computable record and rational $C > 0$, an ordinary rational trader can instead stake its available wealth on each computed next sign. Its wealth is $C2^n$ along that record and becomes zero after an incorrect sign. Thus a computable countermarket is exploitable without bound by another admissible trader. For a known rule Φ as in Proposition 11, compute its return from the actual prior positions before each stake; the same doubling follows. The trader still depends on the specified record or rule. The universal process above uses a different computational class, and success concerns infinite-time unboundedness, not guaranteed profit at a fixed horizon.

a. A probability bound Under independent fair signs, every predictable admissible trader, including one us-

ing an independent private random seed, satisfies Ville's bound [35]:

$$\mathbb{P}\left(\max_{0\le t\le T} G_t \ge K\right) \le \frac{C}{C+K}, \qquad K>0.$$

For randomized traders, require measurable positions and admissibility on every history for each seed. Nonnegativity of both next wealth values gives $\epsilon P_t|w_t| \le C+G_t$, hence $0 \le C+G_t \le 2^t C$. Thus finite-horizon integrability follows automatically. Stopping this martingale when it first reaches $C+K$, or at T, gives $C \ge (C+K)\mathbb{P}(\max_{t\le T} G_t \ge K)$. This quantifies Proposition 7 without requiring computability of the trader.

VIII. RAMSEY REGULARITY AND THE FINITE-PATTERN TRAP

A large market record can be pattern-rich even when it contains no exploitable law. Ramsey theory makes this finite combinatorial point without assuming a probability model. For fixed positive integers c, m, q, there is a finite number $R_q(m;c)$ such that every assignment of one of c colors to each q-element subset of any set of size at least $R_q(m;c)$ has an m-element subset whose q-subsets all receive the same color. Thus every red–blue coloring of the pairs among six objects contains a monochromatic triangle, with all three pairs the same color. Van der Waerden's theorem similarly forces an arithmetic progression—equally spaced indices—of any prescribed finite length, all of one color, in every sufficiently long finite coloring [36, 37]. These thresholds may be enormous, but they are finite and computable, unlike a Busy-Beaver horizon.

The guarantee depends on the coding and is weaker than a trading signal. In a discretized return record, equal symbols at equally spaced dates need not be consecutive, form a price trend, have a favorable sign, or persist beyond the sample. Ramsey existence therefore asserts neither statistical dependence nor causation, predictability, or profit. Nor does it conflict with algorithmic randomness. A fair-coin Martin-Löf-random sequence is Borel normal, so every finite binary word, including arbitrarily long runs and finite chart motifs, occurs with its fair-coin limiting frequency. Randomness here means escaping every effective null test, not local patternlessness [33, 38].

Calude and Longo call regularities forced by data volume *Ramsey-type correlations*. Their incompressibility argument emphasizes that almost all sufficiently long finite strings resist compression by any fixed positive fraction while still containing the finite regularities forced by Ramsey theory [39]. This is an epistemic warning, not a theorem that every sample correlation is false: a Ramsey configuration is not itself a correlation coefficient, and a pattern observed in one finite history is compatible with both random and computable continuations.

Trading research compounds this pattern abundance with selection. A null hypothesis is the baseline claim assessed by a statistical test, such as the absence of a forecasting relation. If all M null hypotheses are true, their rejection events are independent, and each has rejection probability exactly α, the probability of at least one false rejection is $1-(1-\alpha)^M$; if their probabilities are at most α, this is an upper bound. Without independence, the union bound, which bounds the probability of any rejection by the sum of individual rejection probabilities, gives $\min\{1, M\alpha\}$ as an upper bound. This probabilistic multiple-testing problem is distinct from Ramsey's deterministic existence theorem, but its practical lesson is aligned: a trend or cross-market relation retained after searching many assets, windows, transformations, lags, and parameters is a candidate hypothesis, not yet an edge. A defensible FAPP claim must report the search universe and survive selection correction (accounting for the search over alternatives), costs, an untouched holdout (data never used to choose or tune the strategy) or independent replication, and testing outside the selected regime [40].

IX. COUNTERTRADERS AND MUTUAL COUNTERING

Definition 1 (Countertrader)**.** For a specified market and initial price, a *countertrader* is a total deterministic computable strategy satisfying

$$w_t(P_{t+1}-P_t) > 0$$

at every date along its interaction with that market. The strategy may depend on the market's program; the definition does not require discovering an unknown program from observations.

Every fixed positive rational-valued computable price path with a nonzero change at every date admits such a trader. Given its generating program, compute the next change and set $w_t = \operatorname{sgn}(P_{t+1}-P_t)$, where sgn returns $+1$ for a positive argument and -1 for a negative one. The gain is $|P_{t+1}-P_t| > 0$. This is still a strategy of the form used in Section II: it can use the supplied history's length to find the date, compute the specified path, and ignore the observed price values; on the unused empty history, return zero. Its guarantee concerns that path only, and relies on the costless trading and unrestricted computation time assumed above.

A similar construction applies to a market that reacts to earlier positions. The restriction to *earlier* positions specifies the information available when the current return is computed.

Proposition 11 (Countertrader for a Computable Market Rule)**.** *Fix rational $P_0 > 0$ and rational $0 < \epsilon < 1$. Let a total computable function $\Phi\colon \mathbb{Q}^{<\omega} \to \{-\epsilon, +\epsilon\}$ determine the next proportional return by*

$$r_t = \Phi(w_0, \dots, w_{t-1}), \qquad P_{t+1} = P_t(1+r_t).$$

Then this market admits a countertrader, with strictly positive gain at every finite horizon $T \geq 1$.

Proof. Recursively choose $w_t = \operatorname{sgn}\Phi(w_0, \ldots, w_{t-1})$. Only previously computed positions are needed, so totality of Φ makes the position sequence computable. Its gain in period t is $w_t P_t r_t = \epsilon P_t > 0$; summing gives the horizon claim. To obtain a total function on arbitrary nonempty price histories, use their length to regenerate this position sequence; return zero on the empty history. □

This does not contradict Theorem 1. That theorem fixes a trader M_σ and constructs its countermarket P^{M_σ}. A countertrader exploiting this path is a different trader. Constructing a countermarket against the replacement can produce a different path again. Thus every fixed trader has a countermarket, while every fixed computable path with nonzero changes has a countertrader. Each responding program depends on the opponent already specified. The quantifiers express dependence, not the temporal order of deployed trades. The success criteria also differ: the countertrader earns strictly positive gain, whereas the countermarket prevents strictly positive gain and cannot make an abstaining trader lose. Allowing flat price paths would invalidate the stated countertrader guarantee.

a. Perfect current responses Write $a_t \in \{-1, +1\}$ for a unit position and $d_t \in \{-1, +1\}$ for the return direction. A perfectly profitable trader and a perfectly opposing market would require, respectively,

$$a_t = d_t, \qquad d_t = -a_t.$$

These equations have no solution. This rules out their simultaneous satisfaction, not interaction between arbitrary deterministic programs: any two total rules on past histories produce well-defined play. If each program first simulates the other on the same current history and returns only after that simulation returns, neither computation terminates. This violates the totality assumption [41].

If the trader announces its current position before the market chooses the return, the market can oppose every nonzero position; abstention secures zero. If the market commits to a nonzero current return and reveals it before the position is chosen, the trader can profit. Knowledge of a deterministic opponent's code can sometimes replace observation of its action, but does not make incompatible perfect-response requirements consistent. Calling a market rational also requires assigning it an objective, such as minimizing this trader's gain; computability alone supplies no such objective.

Total computability supplies no trading deadline. A simulation may finish too late. In the mutually dependent current-response construction, timeouts or fallback actions can restore termination only by abandoning at least one exact countering requirement. There is no general rule that a counterstrategy needs a larger computational complexity class: an always-up market is countered by an always-long trader. Equal computation budgets alone determine neither the winner nor a zero payoff.

b. Hidden random choices For a simple one-period simultaneous game, fix the current price $P_t > 0$, restrict positions to $w_t \in \{-1, 0, +1\}$, retain directions $d_t \in \{-1, +1\}$, and assign the trader payoff $\epsilon P_t w_t d_t$ and the market its negative. This is matching pennies with an abstention option. Its value is zero. A mixed equilibrium, meaning neither side can improve its expected payoff by changing its own choice distribution alone, has a fair market direction and equal trader probabilities for long and short. Any remaining probability may be assigned to abstention. Choices are hidden and independently randomized. The stated distributions give expected payoff zero against every opposing pure action, which proves the equilibrium claim. There is no pure equilibrium: any fixed nonzero direction can be exploited, and any active fixed position can be opposed.

This equilibrium statement is distinct from Proposition 7. That proposition makes every predictable trader's gain, subject to its integrability assumptions, a martingale under the specified fair-sign market law. It does not require mean-zero positions or establish mutual best responses. For example, always holding one share has expected gain zero under that law, although an opposing market would prefer downward moves.

Reflective-oracle models provide a more general framework for consistent mutual reasoning through oracle access and probabilistic answers [41]. This changes the computational model; ordinary randomization does not make exact universal next-move prediction possible. Related tensions between prediction and optimization arise in repeated-game learning under assumptions on beliefs or uncertainty about opponents' payoffs [42, 43]. These results provide context; the countertrader construction and the incompatible current-response equations already establish the distinction needed here.

X. PRACTICAL ROBUSTNESS DIAGNOSTICS: TIME REVERSAL AND PERMUTATIONS

This section turns the trader–market leitmotif into a falsification tool. Hold the trader fixed and transform only the market record. Among the countable family of computably specified stress transformations, time reversal is one of the simplest, particularly for detecting dependence on directional persistence. It is a diagnostic, rather than a further impossibility theorem like the countermarket construction or Halting reductions.

Let $\mathcal{M}$ be a space of strictly positive discounted price paths $P\colon [0, T] \to \mathbb{R}^n_{>0}$, where $\mathbb{R}^n_{>0}$ denotes vectors of n positive real prices, one per asset, and T is the horizon. The time reversal of P is

$$\widetilde{P}(t) = P(T - t), \qquad t \in [0, T]. \tag{10}$$

Literal reversal reuses the observed price levels and pre-

serves the magnitudes of log-price moves, the logarithms of successive price ratios, while reversing their order and signs. It is not the same as merely negating arithmetic returns: a gross return R is the later price divided by the earlier price, and the corresponding arithmetic return is $R-1$. The reversed gross return is $1/R$. In an empirical implementation the strategy must be rerun causally from scratch on the reversed record, using only observations already available at each decision. Reversing the original orders or realized profit would leak future information. If the initial level is prescribed, literal reversal can leave the market class. One may instead test componentwise normalized reversal,

$$P_i^{\mathrm{rev}}(t) = P_i(0)\frac{P_i(T-t)}{P_i(T)},$$

where i labels the asset and rev its reversed path, provided it remains in the claimed class. Scaling can change the actions of a strategy that is not scale invariant, meaning that its decisions change when all prices of an asset are multiplied by the same positive constant; it is therefore a distinct test. For a causal strategy σ, let $\Pi_t[\sigma; P]$ denote its discounted gain from zero initial capital through time t on P. Call σ *pathwise admissible* on $\mathcal{M}$ if some finite C satisfies $\Pi_t[\sigma; P] \geq -C$ for every $P \in \mathcal{M}$ and $t \in [0, T]$.

Definition 2 (Pathwise Universal Strategy)**.** A causal strategy σ is *pathwise universal* on $\mathcal{M}$ if it is self-financing and pathwise admissible there and satisfies $\Pi_T[\sigma; P] > 0$ for every $P \in \mathcal{M}$.

Criterion 1 (Reversal Falsification Test)**.** Suppose $\mathcal{M}$ is closed under time reversal, meaning that reversing any member produces another member. If there exists $P \in \mathcal{M}$ such that

$$\min\{\Pi_T[\sigma; P], \Pi_T[\sigma; \widetilde{P}]\} \leq 0, \tag{11}$$

then σ is not pathwise universal on $\mathcal{M}$.

Proof. Both P and $\widetilde{P}$ belong to $\mathcal{M}$. A non-positive gain on either member of the pair contradicts Definition 2. □

One non-winning path refutes a claim quantified over every path. Reversal supplies a paired candidate, not a proof that one member must fail: a causal strategy may take different positions on the reversed history. The countermarket construction provides the impossibility result for total deterministic computable traders; reversal is a practical probe of a proposed strategy.

a. Permuting a finite record Time reversal is a special permutation of a sampled record. For a finite record of discounted price levels $P = (P_0, \ldots, P_N)$, let π be a permutation of $\{0, \ldots, N\}$, a rearrangement using every index exactly once, and define

$$(P^\pi)_t = P_{\pi(t)}, \qquad t = 0, \ldots, N. \tag{12}$$

Thus $\pi(t) = N - t$ gives reversal. A fixed algorithm or a random shuffle can choose π. For vector-valued prices, each complete price vector moves together. Let $\mathcal{M}_N$ be the claimed class of such discrete records and $\Pi_N[\sigma; P]$ the discounted gain from zero initial capital when the same causal rule is rerun from its prescribed initial state on P. Use the discrete version of Definition 2 on this class.

Criterion 2 (Permutation Falsification Test)**.** A pathwise universal strategy on $\mathcal{M}_N$ must satisfy

$$\begin{aligned} &\Pi_N[\sigma; P^\pi] > 0 \\ &\quad \text{whenever } P \in \mathcal{M}_N \text{ and } P^\pi \in \mathcal{M}_N. \end{aligned} \tag{13}$$

One permitted permutation with zero or negative gain refutes that universal claim. The criterion requires positivity, not equality of profits across permutations.

Proof. Every retained P^π is a member of the claimed market class, so strict profit on every member requires strict profit on this one. □

The shuffle rearranges price levels, not returns or previously chosen orders. At each date the rerun uses only the history available in the new order. Permuting levels preserves their multiset, the collection of levels with repetitions counted, but generally changes the returns and their magnitudes. It can also violate a fixed initial-price condition, a bound on price moves, or another restriction defining $\mathcal{M}_N$. Only transformations that remain within that class test its universal claim; closure under all permutations is unnecessary. Failure after destroying a trend outside a strategy's stated domain does not refute a claim confined to that domain. Passing finitely many shuffles of one record does not establish universal profitability on a larger path class.

This diagnostic assumes no uniform distribution over records or equality of average performance. Randomness in the choice of stress records alone supplies neither a market law nor a statistical significance level.

Greg Egan's *Permutation City* (1994) [44] provides a literary analogy by exploring the temporal ordering of simulated computational states. Here permutations change the trader's observation sequence; the causal rule is rerun on each record, and Criterion 2 follows from the universal-profit requirement.

Remark 3 (Computable Stress Tests Beyond Reversal). Time reversal is one of countably many computable probes: reversing return signs, joining segments from different market conditions (regime splicing), adding price jumps or execution delays, changing the size of price fluctuations (volatility rescaling), and varying financing or transaction costs. Since program totality is undecidable, an operational battery must select known-terminating transformations whose outputs remain in the claimed market class. Here "test" means empirical stress, not a Martin-Löf test: a uniformly computably enumerable sequence of open sets with effective measure bounds. Reversal defines no null set and certifies no randomness; passing it shows only tested robustness, not universality.

Remark 4 (Filtrations and Time Reversal). Stochastic model classes are not automatically reversal-closed: The forward filtration is the information accumulated in the original time order. The reversed value $P(T-t)$ is generally not adapted to it, meaning that this value need not be determined by information available at time t. Reversing a diffusion, a continuous process driven by Brownian noise, requires additional hypotheses [45]. For example, if

$$X_t = \log(S_t/S_0) = mt + \nu W_t,$$

X_t is the stock's cumulative log return, m its drift (average rate of change), and W_t standard Brownian motion, whose independent normal increments have mean zero and variance equal to elapsed time. Here $\nu \geq 0$ denotes volatility, the scale of random fluctuations. Then $X_{T-t} - X_T$, under the information revealed by the reversed increments, has drift $-m$ and volatility ν. Thus a class containing both drift signs is closed in law under this transformation: the transformed process has a distribution within the class. Under the physical, or real-world, probability law, $m = \mu - \nu^2/2$, where μ is the stock's instantaneous expected proportional return. Under a risk-neutral measure, a probability law used for pricing through discounted expectations, the discounted stock may be a martingale while its logarithm retains the Itô drift $-\nu^2/2$, the correction caused by taking logarithms of Brownian-driven prices. Zero drift in an additive Brownian price model does not establish detailed balance in general, meaning invariance of equilibrium history probabilities under time reversal. The diagnostic remains pathwise.

XI. CONCLUSION

Everything returns to the same pair: a trader and a market. For every fixed total deterministic computable trader, Theorem 1 constructs a fixed positive computable price path whose next bounded proportional move opposes every nonzero position and remains nonzero during inactivity. The quantifiers are $\forall\sigma\,\exists\mathcal{E}_\sigma$: no physical market must observe the deployed orders, the always-flat trader earns zero, and every nonzero position loses on its trader-relative countermarket. This direct diagonal result uses no Halting-Problem reduction and identifies no single path that defeats every trader.

The converse construction is also opponent-specific. A trader supplied with a total computable program for nonzero price changes can choose profitable positions, but remains vulnerable to a new countermarket. These separate constructions do not yield mutually perfect current-move responses: their interaction requires an explicit account of timing, information, and termination (Section IX).

The remaining limits concern different market sides. Gold precludes identifying every computable binary generator in the limit, while next-bit diagonalization defeats each fixed computable predictor on its predictor-relative record. Halting reductions preclude uniform certification of eventual profit or encoded future events, and Busy Beaver excludes a computable description-size waiting horizon despite finitely solvable low-complexity cases. Under a stated computable reference measure, the generated record is Martin-Löf random almost surely; neither this property nor Borel normality can be certified from a finite sample. Ramsey regularity explains why finite structured subsequences nevertheless remain abundant, including in random records. These conclusions are complementary because their inputs, quantifiers, and success criteria differ.

Proposition 3 preserves cumulative nonpositivity even for approximation-based computable-real positions, although a computable path need not make every active position lose. The martingale correspondence in Section VII concerns bounded-credit trading and unbounded wealth over an infinite record. A universal lower-semicomputable capital process therefore does not provide an executable universal trader with strict profit at a prescribed finite horizon.

These limits leave room for FAPP profitability under specified market conditions, lawful information, or compensation for risk. Such claims remain conditional and must be evaluated after costs and financing, against an explicit benchmark. The selection controls in Section VIII and the possibility that deployment changes the market remain essential; neither establishes a universal profit guarantee.

Any repeatable-success claim must therefore state its market restriction, probability law, benchmark, admissibility and financing conditions, horizon, data protocol, and deployment effects. Time reversal and other within-class permutations are robustness probes: one nonpositive gain refutes an all-path claim, while passing a finite set of probes does not certify universality. Useful strategies may exist, but no total deterministic computable self-financing rule with exact rational-valued positions guarantees strict gain on every positive rational-valued computable price path.

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author accepts full responsibility for the final text.

DATA AVAILABILITY

This is a purely mathematical work, and no data or software were created or analyzed in this study.

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